NOTE - READ THIS FIRST: 4 WHEELS IS THE SAFER DESIGN. DUMB LAWS MAKE IT LESS SAFE.

This kart is designed with **4 wheels** because 4 wheels is the safer design for kiddohs - harder to tip, steadier to ride. Due to dumb laws, the result will be less safe than it should be: California law punishes the safer 4-wheeled design, so kiddohs either get a tippier 3-wheeler or get pushed off the bike paths. The law, not the engineering, is what costs safety.

Yell at: the State Assemblymember, District 2, and the State Senator, District 2. Full list, in order: page 3.

DRAFT

PART 1 - WRITE-UP | PART 2 - SYLLABUS

The pedal kart to the next town.

A kiddoh should be able to visit a friend - or a puppy love interest - in the next town without a 3-hour bus ride eating the afternoon.



Illustration only - one possible look for the 4-wheeled kart. Not a photograph.

The problem. Town A to Town B is about 5-15 *miles* (guesstimate range). By bus: up to about 1-3 hours (guesstimate range). By trail, where a separated path exists: one ride of about 20-60 *minutes* (guesstimate range). A class measures its own towns and replaces these numbers (Unit 1).

The answer is not a smaller car. It is a kart built to bicycle limits: 4 wheels, real pedals, just enough electric help to reach the next town - the rider still does the work, and students design the form details. But California law today does not give a 4-wheeler this slow clean bicycle treatment.

> **In one line:** 4-wheeled pedal-assist kart at bicycle limits, form details by the students + swap stations at choke points + difficulties left in on purpose, so the trip teaches.

Bicycle limits, 4 wheels, one dumb conflict

California is the law design target: its laws are among the hardest of any state, so passing California means built to the worst case, and the design travels well to easier states. Not "legal in all 50 states" - each Town A / Town B pair checks its own state and trail rules.

WHAT DOES NOT WORK

- **Low-Speed Vehicle (LSV).** Needs a driver's licence - kiddohs are out.
- **Converted go-kart.** No VIN, no manufacturer certification. Adding pedals does not reclassify it.
- **Class 3 e-bike.** 16+ only, and restricted off many bike paths.

THE DESIGN: 4 WHEELS

- **4 wheels - set by Matthew.** Most stable, hardest to tip. For kiddohs, that is the whole point.
- Fully operable pedals that really propel it.
- Motor **750 W max**, help only while pedalling, stops at **20 mph**, labelled; **no throttle** - pedal, or do not go.
- White front / red rear light, working brakes, helmet under 18.
- Students design the rest: body, steering, drive.

The conflict: under current California law this design does **not** get clean bicycle status. Vehicle Code section 407 treats a motorized quadricycle as a restricted category (physical disability or senior citizens), and e-bike rules are written around fewer wheels - so the 4-wheeler loses clean bike-path access and sits in a grey zone, while the cleanly blessed version is a tippier 3-wheeler. This document keeps the 4-wheeled design, names the law as the problem, and does not pretend a 4-wheeler is a clean Class 1 bicycle today.

TWO KIT STYLES - FLAT PACK FIRST, CUSTOMS LATER

WELDED KITS (CUSTOMS - LATER YEARS)

- **More customisable** - each build shaped to its designer.
- Welding skill + hot-work safety needed; builds vary; repairs need a welder.

FLAT-PACK KITS (ERECTOR STYLE - YEAR 1)

- **More uniform** - identical, interchangeable standard parts; uniformity is what makes mass production work.
- Hand tools only, no hot work - assembly a kiddoh can do. Kits ship flat: the export form.

Phase order - set by Matthew: Year 1 is definitely flat pack - every build uniform, the baseline proven, spares and pack compatibility holding. Over the years it translates into customs: welded builds grow out of the proven kit. You earn customs by proving the uniform kit first; a custom keeping the standard interfaces (pack, charging, safety spec) stays part of the system.

Who to yell at – and the Cali domino

A state-law problem: fixed at state level or not at all. Positions, not names – positions survive elections, names do not.

- **First domino: the State Assemblymember, District 2** (Town A's and Town B's district) – to author one small, narrow amendment: bicycle treatment for a pedal-assist 4-wheeled cycle at bicycle power and speed limits. Everything after only exists once a bill exists.
- **Next: the Chair, Assembly Transportation Committee** – the bill must pass this committee.
- **Next: the State Senator, District 2** – carries it in the other house; its current holder is termed out in December 2026 – the window is short.
- **Next: the Chair, Senate Transportation Committee** – the second committee gate.
- **Last domino: the Governor of California** – signs, or vetoes.

The fix: amend the Vehicle Code (sections 312.5 / 407 area). City council and the county cannot fix state law – only pass resolutions asking the state to act.

KNOCK THE CALI DOMINO

California itself is the first domino. Win here, prove it under the hardest rules, and the package exports – same design, same syllabus, same narrow amendment, each state swapping in its own numbers and positions. Knock the Cali domino – the rest get easier.

The vehicle, plainly: light, geared low, narrow enough for bollards and bridges. Pre-ride check every trip: tyres, brakes, chain, lights, pack. No check, no pack.

NUMBER LEDGER – READ EVERY NUMBER BY ITS KIND

Number	Kind / source
750 W max, 20 mph assist cutoff	Design limits – California Class 1 spec
Town A–B: 5–15 miles; ride 20–60 min; bus 1–3 hours; segments 2–5 miles	Guesstimate ranges – a class measures its own route (Unit 1)
Choke-point stations; difficulties left in; 4 wheels	Set by Matthew – design rules
~45 seconds per swap (Gogoro)	Reported precedent, at scale

Range is infrastructure, not a bigger battery

A bigger battery makes the vehicle heavier, dearer, and harder to pedal. So the pack stays small, and the route carries the range.

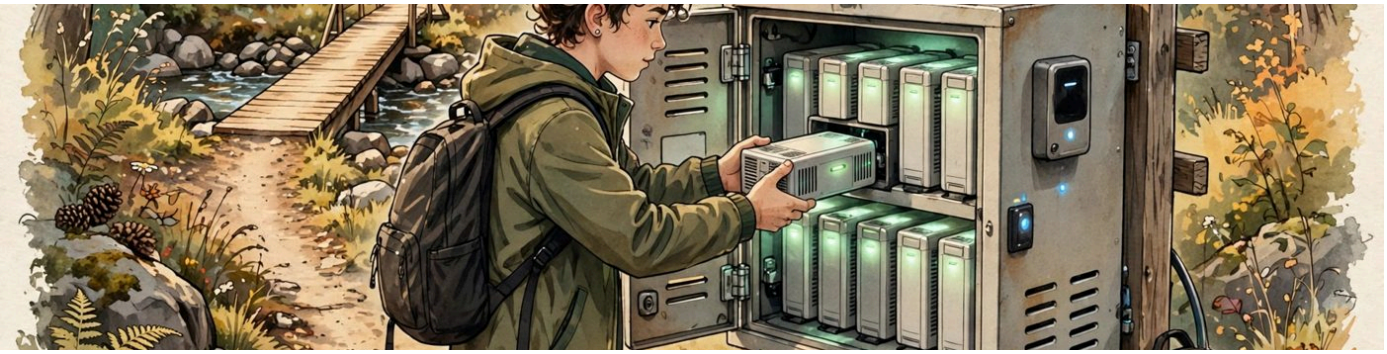


Illustration: a swap at a choke point. Slide the spent module in, take a charged one out.

SPLIT THE PACK IN TWO

Two light modules instead of one heavy pack – about 250–300 watt-hours each (Fermi estimate). A younger kiddoh can lift one in each hand. The swap, not the visit, is the stop – and it takes under a minute.

THE BATTERY IS A COMMONS

- Kiddohs do not own the battery. The system does. That cuts the price of the vehicle and takes the fire-risk part out of bedrooms and garages.
- Stations are monitored, ventilated, fire-safe cabinets, opened by RFID card or fob – no phone, no credit card needed – and every returned module is graded there.
- A damaged pack is quarantined, not handed to the next kiddoh.
- Each pack carries a system identity. A stolen pack is a paperweight.

Placement rule – set by Matthew: not too sparse, and not everywhere. Stations go at choke points only.

Choke points are places everyone has to pass anyway: a bridge, a causeway or gap, a trailhead, a feeder-neighbourhood junction, a main road crossing, a town gate. Segments between chokes: about 2–5 miles (guesstimate range).

Between chokes you are committed to that segment – one segment is the unit of range. At a choke you decide: swap now, or push on? Charge, distance, headwind – that call is the lesson. Town-edge chokes double as a safety reset before traffic.

Friction is the curriculum

Some difficulties are left in on purpose. Not to be mean. Because removing them removes the point.

Difficulty left in	What it teaches
No throttle	Effort is the price of the trip. He rode there. He was not driven there.
Range = one hop, by design	Planning. Check the pack, check the distance, leave with margin.
Stations at chokes only	Resources are placed, shared, and finite. The world is not padded.
No cab, no heater	Judgment. Read the forecast, dress for the wind, decide if today is the day.
The battery is shared	Stewardship. Abuse a pack and the next kiddoh gets the dud - and the system knows it was you.
You maintain your machine	Vehicles are responsibilities, not appliances.

> **The line:** difficulty teaches, danger does not. Every difficulty above is recoverable inside the trip - the worst case is a slow, sweaty ride home on pedals alone. Never left in: darkness without lights, brake or safety faults, fire risk, or being stranded with no pedal fallback.

Pedal either way. Charge it. Slowly.

On the road, forward means go and reverse never propels the vehicle, so the legal form stays clean. Parked, the kart has a second mode: generate. Deploy its own kickstand – a centre stand that props the driven end up, driven wheels off the ground – select the resistance gear, and pedal. Either direction charges: forward pedalling generates, and reverse pedalling generates. A dead pack or an empty station, and you can trickle in enough to limp to the next choke.

No external stand. No separate exercise bike. The traction motor is the generator: the pedals drive it through the resistance gear, it charges the pack, and the resistance in your legs **IS** that generating load. The two directions work the legs differently, so switch when one motion tires the muscles out – two motions provided, so weak muscles are not exhausted by a single repeated one. The controller rectifies either rotation, so direction does not matter electrically. Schools charge swap-pool packs this way in PE, on the kart itself; a rider short on swaps can earn one – about 20–30 minutes of work for range (proposal, pricing not set).

The design lesson: a motor run backwards is a generator – use the hardware already on the vehicle. Generation is a mode, not an accessory; a separate stand is extra parts solving a problem the motor already solves. One honest requirement: the drivetrain must be coupled so pedalling can back-drive the motor in **BOTH** directions, through the resistance gear – a plain freewheel that disconnects pedals from motor breaks generate mode. Students design that coupling in Units 5 and 7.

Human power, honestly	Kind
Fit person: 100–150 W sustained; about 70–120 Wh banked per hour after losses	Reported pedal-generator figures
A kiddoh: about 35–70 Wh banked per hour	Fermi estimate
One hour of legs = about one segment between chokes	Derived from the two rows above
A full ~500 Wh pack by pedalling alone: 7+ hours	Derived

The slowness is the lesson: after an hour for one segment, nobody thinks electricity is free or abstract again. Keep the generating load reasonable in both directions, and switch direction when one motion tires you – that is what the second motion is for.

What is not set yet

Honest list. These are open, not hidden.

- **Kit phasing – SET:** year 1 is flat-pack (erector-style) kits only; welded customs come in later years, grown out of the proven kit. Still open: the exact year or trigger for opening customs.
- **Vehicle form details** – body, steering, drive layout: student design work, not set. Wheel count is set: 4 (page 1).
- **The law change** – not done; until the state acts, no clean bicycle status (pages 2–3).
- **Vehicle name and age range** for the class and for riders – not set.
- **Exact pack interface standard** – not set. One module for every vehicle, forever, or the network fragments.
- **Pricing and first fleet size** – not set. Fleet figures so far are Fermi estimates only: ~20 vehicles, ~70 modules with station stock.
- **Station operator and site permissions** – not set. Sites are mostly somebody's land, bridge, or trail; permission not yet sought.
- **Adult supervision rules** – to be decided before any student ride (see Part 2, Safety).

> Part 2 is the class that builds it, tests it, and runs it.

PART 2 – SYLLABUS

Stuff

How actual stuff works. Learned by building a real system, end to end.

Few words. Do things. Real stuff, real consequences. In Stuff, the project is not a model of a transport system. It is a transport system – small, legal, and ridden by its builders to a real next town.

Proposed shape (proposal): a 12-week term, 2–3 sessions per week. Each unit ends in something a student can show, measure, or ride.

LEARNING OBJECTIVES – BY THE END, A STUDENT CAN

- Read a map for how people move and find its choke points; read a vehicle code as a design brief.
- Measure energy – their own, a battery's, a trip's – in watts and watt-hours, honestly.
- Design the form details around its 4 wheels, then build, check, and maintain it; design shared infrastructure that survives real users, and own the trip plan and the choke-point call.

Units 1–6: problem, law, energy, vehicle

1. The problem and the map. **Do:** map your own Town A and Town B; time the bus; ride or walk the trail; measure the real distance, ride time, and segment lengths, replace the guesstimate ranges with your measured numbers, and mark every choke point. **Lesson:** transport is geography first – and a guesstimate is a placeholder, not a fact. **Assessed by:** a route survey the class can defend – measured numbers in, choke points argued for.

2. The law. **Do:** read the vehicle classes; work out why a go-kart cannot, why an **LSV** shuts kiddohs out, and why section 407 restricts the safer 4-wheeled design. California law is the study target even with generic towns: among the hardest anywhere, so a design that passes it travels. Name the positions that can change it, and the first domino. **Lesson:** the law is a design brief, not an obstacle course – and sometimes the brief is wrong and gets yelled at. **Assessed by:** a one-page spec, in the student's own words, that a builder could build from.

3. Energy I – human power. **Do:** prop the kart on its own centre stand, select the resistance gear, pedal in generate mode, and measure your own output – watts now, watt-hours over an hour. **Lesson:** energy has a body cost. **Assessed by:** the student's own measured data, logged honestly.

4. Energy II – batteries. **Do:** charge and discharge a module; plot the curve; learn slow charging, damaged-pack fire risk, and certified-pack rules. **Lesson:** a battery is stored work with limits. **Assessed by:** a correct charge log and a safety check done right.

5. The vehicle – they design it. **Do:** design the form details yourselves – body, steering, drive layout – around 4 wheels, defend the choices (stability, trail width, cost), then build it: drivetrain, gearing, brakes, lights, real pedals. Drivetrain requirement: pedalling must back-drive the motor in both directions through the resistance gear – a plain freewheel that disconnects pedals from motor breaks generate mode. Year-1 classes build the flat-pack kit; welded custom work belongs to later-year, advanced classes. A mass-production design must still survive being built identically by other hands from the parts list alone. **Lesson:** every choice is a trade-off you have to argue; if a part is decorative, it is lying. **Assessed by:** a design defence, an unassisted pedal test, and a full pre-ride check.

6. Pedal-assist electronics. **Do:** torque and cadence sensing; set the 750 W / 20 mph limits; test what an unlock switch does to the legal category. **Lesson:** a faster setting can delete the whole vehicle, legally. **Assessed by:** a measured top-assist run at or under the limit.

Units 7–12: generation, infrastructure, the run

7. Generate mode – charge by pedalling. **Do:** deploy the kart's own centre stand so the driven wheels are up, select the resistance gear, and pedal for an hour – forward or reverse, both charge, because the motor is the generator and the controller rectifies either rotation; no external stand or rig. Count the watt-hours and convert them to miles. **Lesson:** a motor run backwards is a generator – generation is a mode, not an accessory; alternating motions is load-sharing for muscles, the body's version of not running one part until it fails; and efficiency losses are real – one hour of legs buys about one segment. **Assessed by:** the lab record – energy in, energy banked, losses explained.

8. Infrastructure – the swap station. **Do:** inventory maths; model the after-school surge; lay out a fire-safe cabinet, RFID access, and pack grading. **Lesson:** a station that is empty at 3:30 pm is an empty box with a line at it. **Assessed by:** a station plan sized for the surge, not the average.

9. The commons. **Do:** run the shared pack pool for a week; track who returns what; design the earn-a-swap rule. **Lesson:** shared things survive on stewardship, not hope. **Assessed by:** pool records kept clean, and a fairness rule the class will live under.

10. Planning and judgment. **Do:** plan real trips – charge, distance, headwind, daylight, weather; practise the choke-point call: swap or push on. **Lesson:** judgment is a skill, and it is trained by deciding. **Assessed by:** a written trip plan that holds up on the day.

11. Build, test, fail. **Do:** prototype trials; poke holes on purpose; safety review; fail recoverably and log it. **Lesson:** a failure you planned for teaches; a failure you hid teaches nothing. **Assessed by:** a failure log with causes and fixes, not excuses.

12. The run and the review. **Do:** the full inter-town trip, Town A to Town B, station to station. Then review the data and write it up. **Lesson:** the trip is the test. **Assessed by:** completing the run under the rules, plus each student's own write-up of what the difficulties taught them.

> **Assessment, throughout:** demonstration and data first. A student shows it works, shows the numbers, or rides it. Written work explains what the hands already did.

Materials, safety, grading

MATERIALS AND EQUIPMENT (PROPOSAL)

- Frames, wheels, and chassis materials for the student-designed 4-wheeled kart(s) (year 1: flat-pack kit form) – each kart with its own centre stand and resistance gear, so generate mode needs no external stand or separate generator rig.
- Split battery modules to the class standard (once set), chargers, a monitored fire-safe charging/swap cabinet, RFID cards/fobs.
- Power meter / watt-hour logging, basic tools, tyre and brake spares, helmets for every rider, front and rear lights, locks.
- Maps of the class's own Town A to Town B route (surveyed in Unit 1), weather access, and a class logbook – paper counts.

SAFETY – NOT NEGOTIABLE, NOT A LESSON

- Helmet for every rider under 18, every ride. Lights on after dark – no exceptions, no lesson version.
- No vehicle rides with a brake, steering, or battery fault. A fault grounds the vehicle until it is fixed and re-checked.
- Batteries charge only in the monitored cabinet. Damaged packs are quarantined, never ridden, never taken home.
- Adult supervision rules for road and trail sessions: **to be decided before the first ride.**
- Every difficulty in this course is recoverable. If a planned difficulty stops being recoverable, it stops being in the course.

GRADING (PROPOSAL)

Pass by demonstrated competence, unit by unit. A student passes a unit when they have done the thing, shown the data or the working result, and can explain it in plain words. No unit is passed on attendance alone, and the final run (Unit 12) requires Units 2, 4, 5, and 10 first – law, battery safety, vehicle check, trip planning: gates, not options.

WHAT STUFF IS

Stuff is not about stuff in the abstract. It is: pick a real thing people need. Find out what the law, the physics, and the map say. Build the smallest honest version. Leave the hard parts in where the hard parts teach. Run it. Write down what happened.

The pedal kart is the first project. It will not be the last thing Stuff builds – but every later project gets the same treatment: real stuff, real consequences, few words.

DRAFT – Set by Matthew: the purpose, the choke-point rule, the deliberate difficulties, generate mode by pedalling, either direction; the design is 4-wheeled – law, yell-at list, Town A / Town B, and guesstimates as labelled. Proposals (labelled as such in the text): the 12-week shape, unit order, earn-a-swap timing, materials list, and grading approach. Estimates are labelled Fermi where they appear. Nothing in this draft sets price, age range, fleet size, or site permissions.